

Clinical Paper

Myocardial infarction is a frequent cause of exercise-related resuscitated out-of-hospital cardiac arrest in a general non-athletic population



Helle Søholm^{a,*}, Jesper Kjaergaard^a, Jakob Hartvig Thomsen^a, John Bro-Jeppesen^a, Freddy K. Lippert^b, Lars Køber^a, Michael Wanscher^c, Christian Hassager^a

^a Department of Cardiology 2142, The Heart Centre, Copenhagen University Hospital Rigshospitalet, Denmark

^b Emergency Medical Services, Copenhagen, The Capital Region of Denmark, Denmark

^c Department of Cardiothoracic Anesthesia 4142, The Heart Centre, Copenhagen University Hospital Rigshospitalet, Denmark

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ABSTRACT

Background: Performing exercise is shown to prevent cardiovascular disease, but the risk of an out-of-hospital cardiac arrest (OHCA) is temporarily increased during strenuous activity. We examined the etiology and outcome after successfully resuscitated OHCA during exercise in a general non-athletic population.

Methods: Consecutive patients with OHCA were admitted with return of spontaneous circulation (ROSC) or on-going resuscitation at hospital arrival (2002–2011). Patient charts were reviewed for post-resuscitation data. Exercise was defined as moderate/vigorous physical activity.

Results: A total of 1393 OHCA-patients were included with 91(7%) arrests occurring during exercise. Exercise-related OHCA-patients were younger (60 ± 13 vs. 65 ± 15 , $p < 0.001$) and predominantly male (96% vs. 69%, $p < 0.001$). The arrest was more frequently witnessed (94% vs. 86%, $p = 0.02$), bystander CPR was more often performed (88% vs. 54%, $p < 0.001$), time to ROSC was shorter (12 min (IQR: 5–19) vs. 15 (9–22), $p = 0.007$) and the primary rhythm was more frequently shock-able (91% vs. 49%, $p < 0.001$) compared to non-exercise patients. Cardiac etiology was the predominant cause of OHCA in both exercise and non-exercise patients (97% vs. 80%, $p < 0.001$) and acute coronary syndrome was more frequent among exercise patients (59% vs. 38%, $p < 0.001$). One-year mortality was 25% vs. 65% ($p < 0.001$), and exercise was even after adjustment associated with a significantly lower mortality (HR = 0.40 (95%CI: 0.23–0.72), $p = 0.002$).

Conclusions: OHCA occurring during exercise was associated with a significantly lower mortality in successfully resuscitated patients even after adjusting for confounding factors. Acute coronary syndrome was more common among exercise-related cardiac arrest patients.

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1. Introduction

Although the beneficial effects of regular exercise have been proven for both secondary and primary prevention of cardiovascular disease, it is well known that the risk of an acute cardiac event is temporarily increased during strenuous physical activity – termed the paradox of exercise.^{1–3} Sudden cardiac death is tragic and cardiac arrests occurring during exercise and sport activities frequently draw attention. The extensive attention of exercise-related

cardiac arrest and the following concern may refrain some from exercising. Survival after out-of-hospital cardiac arrest (OHCA) has increased in recent years, however OHCA remains associated with a poor prognosis.⁴ Most current studies on OHCA during sport and exercise focus on sudden death in young professional athletes and the discussion of screening prior to participation in elite sports is ongoing in many countries.^{5–8}

Myocardial infarction is the most common cause of malignant arrhythmias leading to cardiac arrest and minimizing the delay from symptom onset to first medical contact in the pre-hospital setting is recognized as a major factor for outcome.^{9,10} The establishment of emergency medical teams educated in basic life support (BLS) and equipped with an AED in sport stadiums and athletic settings have been suggested to improve fast response

* Corresponding author at: Department of Cardiology 2142, Copenhagen University Hospital Rigshospitalet, 9 Blegdamsvej, 2100 Copenhagen, Denmark.
E-mail address: helle.soholm@gmail.com (H. Søholm).

in case of an acute cardiac event as a benefit for both spectators and athletes.^{11–15} Risk factors such as prior cardiovascular events, smoking, and a recent episode of fatigue or flu-like symptoms are associated with an increased risk of an acute cardiac event, however the association between exercise and susceptibility of a fatal myocardial infarction has not been thoroughly investigated.^{16–20}

We aimed at reporting a large urban experience with outcome and prognosis after resuscitated OHCA occurring in relation to exercise in the general non-athletic population. Furthermore, we investigated the incidence of acute coronary syndrome as a cause of the arrest in both patients with exercise- and non-exercise-related cardiac arrests.

2. Materials and methods

2.1. Patients and study area

From June 2002 through 2011 consecutive patients (≥ 18 years) with OHCA of all causes in the greater Copenhagen area and emergency medical services (EMS) dispatched and attended were included in the cohort study. Pre-hospital resuscitation was attempted in all included patients, and if treatment was terminated before hospital admission the patients were excluded from the study.²¹ Patients with return-of-spontaneous circulation (ROSC) or ongoing CPR at hospital admission were included. Patients with obvious signs of death (e.g. rigor/livor mortis) were excluded, as were non-Danish residents due to unavailable outcome data.

The EMS in Copenhagen, the capital of Denmark, covers the Copenhagen area of 675 km² (260 Mi²), which is inhabited by approximately 1.2 million people. The standard response to cardiac arrest is an emergency ambulance with BLS equipment, a defibrillator and a mobile emergency care unit staffed with a physician (anesthesiologist) and a paramedic. The EMS was dispatched to all patients with presumed OHCA with the treatment protocol adherent to the advanced life support guidelines from the European Resuscitation Council.^{22,23} The attending physician used an Utstein registration sheet as documentation and data were entered in the OHCA-database immediately after end of each case.^{24–26} Patients were admitted for post-resuscitation care at one of eight hospitals in the greater Copenhagen area. The OHCA incidence was calculated as number of arrests per year divided with number of citizens in the EMS coverage area.

A single investigator comprehensively reviewed the individual patient charts with definition of registered parameters prior to chart review. Included variables were pre-arrest co-morbidity assessed by the Charlson Comorbidity Index that takes the severity of 22 conditions into account and is used to predict short-term mortality.^{27,28} Moreover data regarding in-hospital post-resuscitation care, place of arrest (public/non-public) and whether the OHCA-patient were exercising at time of OHCA were recorded. Exercise-related OHCA was defined as OHCA occurring during or within 15 min of cessation of moderate to vigorous physical activity. Activities such as lawn mowing, housekeeping and sexual intercourse were not defined as exercise. The etiology of the arrest was determined after review of the full disease course, and cardiac etiology was defined as acute coronary syndrome (ACS), primary arrhythmia, aortic stenosis or cardiomyopathy (dilated or hypertrophic). ACS was defined as ST-segment elevation myocardial infarction (STEMI) with ST-segment elevation in two contiguous electrocardiogram leads of >0.1 mV in V4–V6 or limb leads, or >0.2 mV in lead V1–V3 or non-ST-segment elevation myocardial infarction (NSTEMI) with or without changes in the electrocardiogram. If cardiac biomarkers or the coronary angiography did not confirm the diagnosis, patients were not categorized as

having ACS.²⁹ The diagnosis of ACS was validated in 100 patients in the cohort by a second investigator blinded to previous assessment.

The regional ethics committee waived informed consent to the study with the reference number: H-2-2012-053, and the study was approved by the Danish Data Protection Agency.

2.2. Outcome

The primary endpoint of this study was survival to discharge from hospital and after 1 year. Patients were followed up to May 1st, 2013 with a maximum follow-up time of 11.1 years and with 99.6% completion. Patients who survived the OHCA but afterwards emigrated were censored at time of emigration ($n = 6$). Mortality data were acquired from the Danish National Patient Registry, which holds vital status on all Danish citizens, all of whom are registered by a unique personal identification number. Neurological outcome was assessed with the Cerebral Performance Category score (CPC) at hospital discharge, and CPC 1 or 2 were categorized as favorable neurological outcome, 3 or 4 were categorized as non-favorable and 5 as dead.²⁴

2.3. Statistics

Continuous variables are presented as mean $\pm$ standard deviation for normally distributed data and as median and interquartile ranges (IQR) for non-normal distributed data. Differences were analyzed with Student's unpaired *t*-test or Wilcoxon rank sum test as appropriate. Categorical variables are presented as number (*n*) and percent and differences are analyzed with χ^2 -test. Kappa statistics were applied for inter-observatory agreement.

Mortality is presented as Kaplan–Meier plots and differences are tested using log-rank test. Univariate and multiple proportional hazard regression analysis (COX-regression) were performed estimating hazard ratios (HR) and 95% confidence intervals (CI) adjusting for potential confounders (age, sex, rhythm, witnessed arrest, bystander CPR, time to ROSC, initiated therapeutic hypothermia, acute coronary angiography, cardiac etiology, public arrest, Charlson Comorbidity Index and arrest during nighttime (midnight to 8 AM) after checking for the underlying assumptions of proportionality and lack of interactions. All statistical analysis was carried out in SAS Statistics version 9.3 (Cary, NC, USA) with a level of significance defined as $p < 0.05$.

3. Results

3.1. Patient characteristics

In the study period a total of 1393 consecutive patients suffered from OHCA, and were attempted resuscitated by the EMS in Copenhagen and admitted to hospital with ROSC or ongoing CPR (Fig. 1) (incidence approximately 0.01% per year). Of those were 7% ($n = 91$) of the arrests occurring during exercise (Fig. 1). Mean age was 60 ± 13 years for OHCA during exercise and 65 ± 15 years in non-exercise OHCA-patients ($p < 0.001$) with no differences in the number of patients younger than 40 years (Table 1). Patients experiencing OHCA during exercise were almost exclusively male (96% vs. 69%, $p < 0.001$) (Table 1). Charlson co-morbidity index was significantly lower in patients with exercise-related arrests ($p < 0.001$), however risk factors for an acute cardiac event (smoking, hypercholesterolemia, known ischemic heart disease, congestive heart failure and/or hypertension) were not different in the two groups (Table 1).

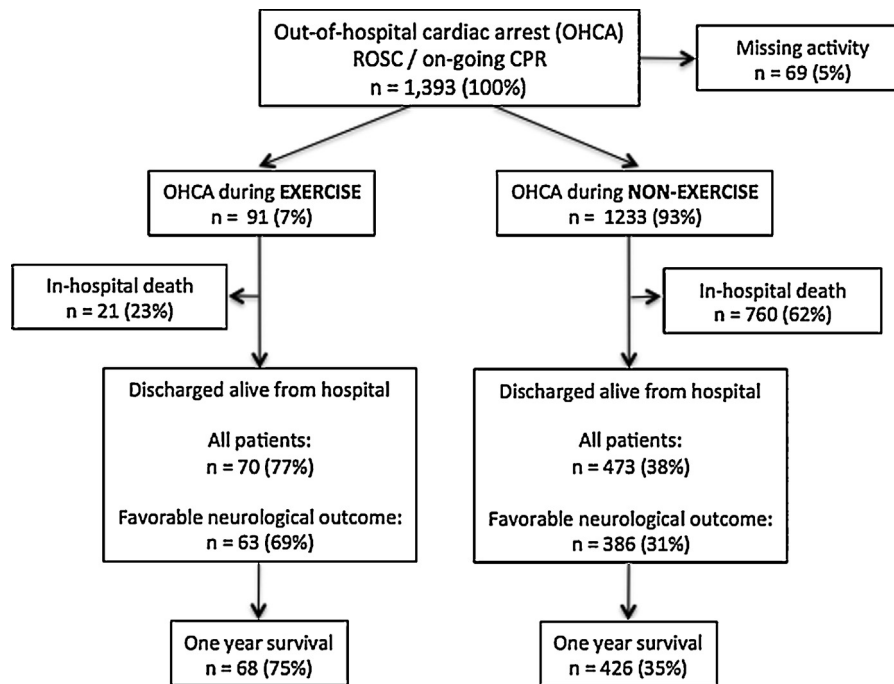


Fig. 1. Flowchart of included patients. Favorable neurological outcome is defines as Cerebral Performance Category (CPC) of 1 or 2 at hospital discharge. OHCA: out-of-hospital cardiac arrest, ROSC: return of spontaneous circulation, CPR: cardiopulmonary resuscitation.

Table 1

Demographic, pre-hospital and post-resuscitation data for patients resuscitated from out-of-hospital cardiac arrest (OHCA) during exercise and non-exercise-related cardiac arrests.

	OHCA during exercise n = 91 (7%)	OHCA non-exercise n = 1233 (93%)	p-Value
Men	87 (96%)	848 (69%)	<0.001
Age (years)	60 ± 13	65 ± 15	<0.001
Men	60 ± 13	63 ± 14	0.03
Women	52 ± 19	69 ± 15	0.03
Public arrest	88 (97%)	401 (33%)	<0.001
Witnessed cardiac arrest	88 (94%)	1016 (86%)	0.02
Bystander CPR	74 (88%)	619 (54%)	<0.001
On-going CPR	9 (10%)	95 (8%)	0.4
Alarm to EMS arrival, min.	7 (5–10)	7 (5–9)	0.3
Arrest after arrival of EMS	0	69 (8%)	0.02
Primary rhythm			
VF/pVT	83 (91%)	604 (49%)	
Asystole	3 (3%)	271 (22%)	<0.001
PEA	2 (2%)	246 (20%)	
Other	3 (3%)	112 (9%)	
Alarm to DC (min)	4 (1–8)	5 (2–9)	0.3
Time to ROSC (min)	12 (5–19)	15 (9–22)	0.007
Acute coronary angiography	38 (42%)	332 (27%)	0.002
Therapeutic hypothermia	53 (59%)	579 (48%)	0.04
Etiology			
Cardiac	88 (97%)	981 (80%)	<0.001
Acute coronary syndrome	53 (59%)	474 (39%)	<0.001
Premature ACS (<50 years)	7 (13%)	44 (9%)	0.3
STEMI	26 (29%)	273 (23%)	0.1
Charlson co-morbidity index	0 (0–1)	1 (0–2)	<0.001
Pre-arrest co-morbidity			
Ischemic heart disease	18 (20%)	256 (21%)	0.9
Smoking	23 (26%)	312 (25%)	0.9
Hypercholesterolemia	24 (27%)	220 (18%)	0.07
Hypertension	35 (39%)	439 (36%)	0.6
Congestive heart failure	10 (11%)	194 (16%)	0.3

Abbreviations: CPR: cardiopulmonary resuscitation, EMS: emergency medical service, VF: ventricular fibrillation, pVT: pulse less ventricular tachycardia, PEA: pulseless electrical activity, DC: defibrillation (direct current), ROSC: return of spontaneous circulation, ACS: acute coronary syndrome, STEMI: ST-segment elevation myocardial infarction.

Bold values are statistically significant.

3.2. Pre-hospital factors and use of AED

Patients with OHCA during exercise had shorter time to ROSC (12 min (IQR: 5–19) vs. 15 (9–22), $p=0.007$) and the primary rhythm was more frequently shock-able (91% vs. 49%, $p<0.001$) compared to non-exercise patients. Patients suffering from OHCA during exercise were in 97% ($n=88$) of cases experiencing cardiac arrest in a public place with 43% of cases ($n=36$) occurring in gyms or sport stadiums, compared with 33% ($n=403$) of non-exercise patients arresting in public, $p<0.001$. Exercise-related OHCA was more frequently witnessed (94% vs. 86%, $p=0.02$) and bystander CPR was more often performed (88% vs. 54%, $p<0.001$) compared to non-exercise-related OHCA-patients.

Automated external defibrillators were used in 24% ($n=19$) of patients suffering from OHCA during exercise and 18 out of 19 patients were treated with the AED after 2008. We found no difference in the number of arrest occurring during daytime (8 AM to 4 PM) (47% vs. 48%, $p=0.8$), however the numbers of exercise-related arrests were significantly higher during the evening (4 PM to midnight) (48% vs. 38%, $p=0.048$) and significantly lower during nighttime compared with non-exercise (midnight to 8 am) (4% vs. 14%, $p<0.001$).

3.3. Etiology of the arrest

Cardiac etiology was the predominant cause of OHCA in both exercise and non-exercise patients (97% vs. 80%, $p<0.001$) and acute coronary syndrome was found significantly more often among exercise-related OHCA patients (59% vs. 38%, $p<0.001$). Inter-observer reliability analysis showed excellent agreement of the ACS diagnosis (kappa=0.96, 95% CI: 0.90–1.00). In OHCA-patients with ACS no significant difference in the proportion of STEMI vs. NSTEMI in the two groups were found (Table 1). Premature coronary artery disease (debut <50 years) did not differ between exercise- and non-exercise-related arrests (13% vs. 9%, $p=0.4$) (Table 1). Primary arrhythmia was the second most common cause of exercise-related arrest, while dilated and ischemic

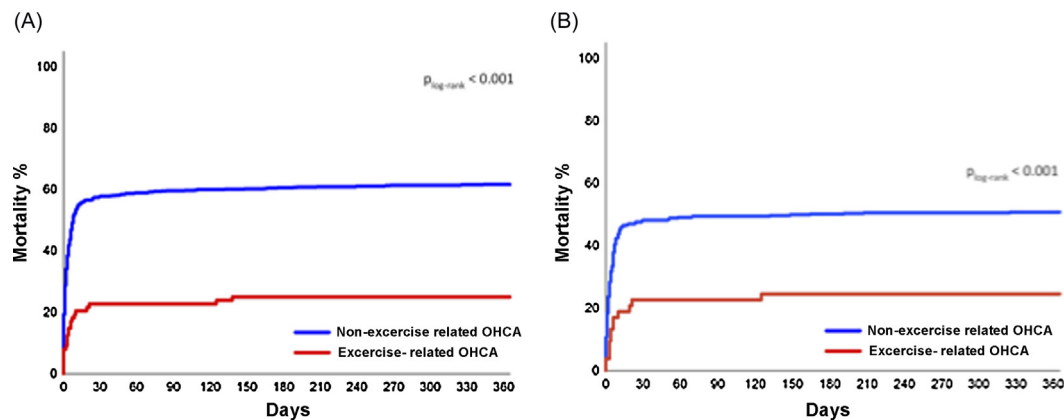


Fig. 2. (A) Kaplan–Meier 1-year mortality plot for all patients with cardiac etiology as a cause of the out-of-hospital cardiac arrests in exercise (red) vs. non-exercise (blue). (B) Kaplan–Meier 1-year mortality plot for out-of-hospital cardiac arrests with acute coronary syndrome in exercise (red) vs. non-exercise (blue). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

cardiomyopathy, severe aortic stenosis, spasm angina, respiratory insufficiency and hypertrophic cardiomyopathy were less common.

3.4. Outcome

In-hospital mortality was significantly different in the two groups with 23% vs. 62%, $p < 0.001$ dying before discharge, for patients suffering from OHCA during exercise vs. non-exercise (Fig. 1). One-year mortality was 25% for exercise-related vs. 66% for non-exercise-related arrests ($p < 0.001$) (Fig. 2) corresponding to an unadjusted hazard ratio (HR) of 0.28 (95% CI: 0.19–0.43), $p < 0.001$ for exercise (Table 2). After adjustment for potential confounders and known prognostic factors, exercise remained associated with a significantly more favorable prognosis (HR = 0.40 (95% CI: 0.23–0.72), $p = 0.002$) (Table 2).

Multivariate interaction analysis was performed including potential confounding factors shown in Table 2 and neurological status at hospital discharge. A significant interaction is found for OHCA during exercise and age ($p = 0.01$) and time to ROSC ($p = 0.02$). No interactions were found with the remaining parameters.

In patients with cardiac etiology of OHCA, 1-year mortality was 25% vs. 62% in exercise vs. non-exercise-related arrests ($p < 0.001$) and the corresponding adjusted HR for exercise was 0.45 (0.24–0.84), $p = 0.01$. One-year mortality in patients with acute coronary syndrome was 25% vs. 51% ($p < 0.001$) in exercise vs. non-exercise-related arrest patients (Fig. 2) corresponding to an adjusted HR for exercise of 0.71 (0.33–1.51), $p = 0.4$.

Further subgroup analysis of arrest during daytime, witnessed arrest, bystander CPR performed, shockable primary rhythm, patients younger than 50 years old and male gender showed that the HR for exercise-related OHCA did not change significantly (supplementary table). In patients older than 70 years old exercise was no longer associated with prognosis (HR = 0.58 (0.23–1.45), $p = 0.2$).

Supplementary table related to this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.resuscitation.2014.06.033>.

A favorable neurological outcome (CPC 1 or 2) in survivors to discharge from hospital was significantly more common among exercise-related OHCA patients compared with non-exercise patients ($n = 63$ (90%) vs. $n = 386$ (82%), $p = 0.04$).

3.5. Exercise

Questionnaires addressed to the general population has previously estimated that Danes who exercise regularly on average

spend 4 h per week on exercise,^{30,31} which corresponds to 2% of the time spend on average on exercise (4 h/(24 h × 7 days)) or 4% of our woken hours (4 h/(16 h × 7 days)).^{30,31} This should be weighed against the 7% (95% CI: 5–8%) overall incidence of cardiac arrests occurring during exercise and resulting in ROSC or ongoing CPR at hospital arrival. In Fig. 3, where the proportion of the general Danish population exercising regularly is depicted by age groups, it is seen that almost 50% of the younger Danish population exercises regularly, which declines linearly with age to less than 10% in the age group >75 years. However, it is noted, that the time spend on exercise in the different groups that exercises regularly does not decline significantly with age (Fig. 3). From the ratio of exercise vs. non-exercise-related OHCA (Fig. 3) it is noted, that the proportion of exercise-related OHCA does decline with age, with 15% (4/26) (95% CI: 6–24%) of the OHCA in the age group 25–34 years being exercise-related and in the population older than 75 years only 2% (8/376) (95% CI: 1–4%) of the arrests occurred during exercise. Bicycling, running and soccer were the most frequent exercise activities in resuscitated victims experiencing cardiac arrest in relation to exercise (Fig. 4). One out of the 91 (1.1%) exercise-related OHCA-patients was a professional athlete (soccer) at time of the cardiac arrest.

4. Discussion

Exercise-related OHCA represented 7% of our consecutive cohort and was associated with a significantly lower mortality rate compared to OHCA occurring during other activities with a 1-year mortality of 25% vs. 65%. Exercise-related OHCA-patients presented with more favorable pre-hospital circumstances with bystander CPR and witnessed arrests being more common. ACS was identified as the most common cause of exercise-related OHCA, however pre-arrest risk factors for development of a sudden cardiac event such as hypertension and hypercholesterolemia was not found to differ between exercise and non-exercise-related OHCA-patients. Even after adjustment for potential risk factors such as pre-hospital factors and co-morbidity OHCA during exercise was associated with a significantly lower mortality rate.

The incidence of exercise-related cardiac arrests vary between 3 and 5% in prior studies and have likewise been associated with higher survival than arrests occurring during other activities.^{17,18} However, in this study we found markedly higher survival to discharge rates in exercise-related arrests (77%) compared with a recent French (15.7%) and Dutch study (46.2%).^{18,19} Patients with termination of treatment in the pre-hospital setting were included in the French study, whereas in the Dutch study only patients

Table 2
Univariate and multivariable predictors of *one-year mortality* for successfully resuscitated patients (ROSC or ongoing CPR at hospital arrival) suffering from out-of-hospital cardiac arrest.

	Univariate		Multivariate	
	HR (95% CI)	p-Value	HR (95% CI)	p-Value
Arrest related to exercise	0.28 (0.19–0.43)	<0.001	0.40 (0.23–0.72)	0.002
Age at arrest (per 5 years)	1.14 (1.11–1.17)	<0.001	1.12 (1.08–1.16)	<0.001
Gender (male)	0.62 (0.54–0.71)	<0.001		
Shockable primary rhythm	0.38 (0.33–0.43)	<0.001	0.44 (0.36–0.54)	<0.001
Witnessed cardiac arrest	0.72 (0.60–0.87)	<0.001	0.76 (0.59–0.97)	0.03
Bystander CPR	0.57 (0.50–0.65)	<0.001	0.72 (0.60–0.86)	<0.001
Time to ROSC (per 5 min)	1.07 (1.03–1.10)	<0.001	1.12 (1.08–1.16)	<0.001
Therapeutic hypothermia	0.52 (0.46–0.60)	<0.001	0.72 (0.59–0.86)	<0.001
Acute CAG	0.36 (0.30–0.44)	<0.001	0.66 (0.51–0.85)	<0.001
Cardiac etiology	0.63 (0.53–0.73)	<0.001		
Public arrest	0.54 (0.47–0.63)	<0.001		
Charlson co-morbidity index	1.31 (1.24–1.39)	<0.001	1.15 (1.06–1.25)	<0.001
Arrest during nighttime (00–08 AM)	1.13 (0.92–1.38)	0.3		

Gender, cardiac etiology and arrest during nighttime are eliminated by backward selection.

Abbreviations: HR: hazard ratio, CPR: cardiopulmonary resuscitation, ROSC: return of spontaneous circulation, CAG: coronary angiogram.

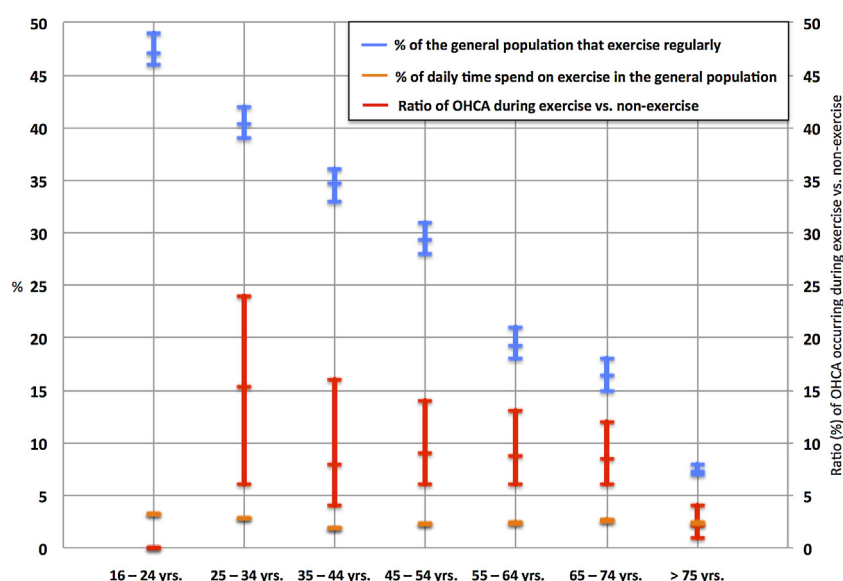


Fig. 3. Proportion of the general Danish population that exercises regularly in their daily life (blue) and percent of daily time spend on exercise (orange) divided by age groups (deciles) with 95% confidence intervals.³⁰ Ratio of out-of-hospital cardiac arrests (OHCA) occurring during exercise vs. other activities (non-exercise) (red) divided by age groups (deciles) with 95% confidence intervals. Exercise-related OHCA was defined as OHCA occurring during or within 15 min of cessation of moderate to vigorous physical activity. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

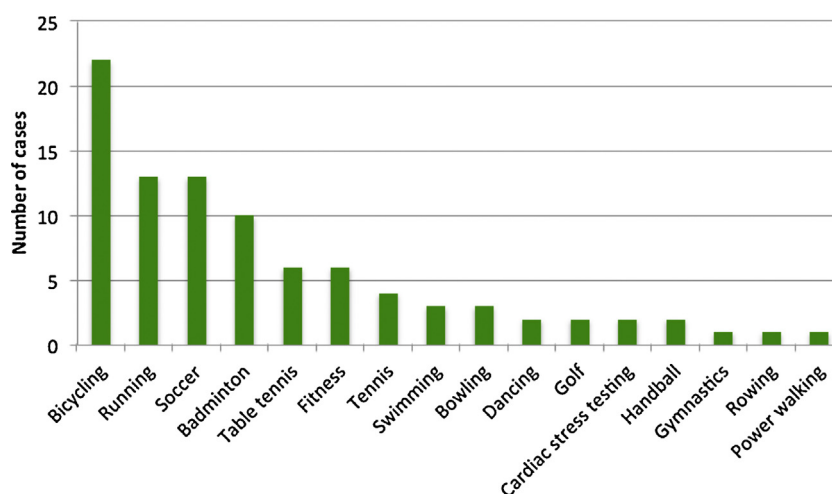


Fig. 4. Exercise activities in cardiac arrest victims experiencing cardiac arrest in relation to exercise.

resuscitated from cardiac arrest were included, similarly to our study.^{18,19} Patients in the French cohort were younger with a mean age of 46 compared with 59 and 60 years in the Dutch and present cohort, respectively.^{18,19} The higher survival rates overall in exercise-related cardiac arrests may partly be explained by more favorable circumstances (more often public and witnessed arrest, bystander CPR and shockable rhythm). In this study interactions between exercise-related OHCA and age and time to ROSC were found. Exercise-related OHCA was associated differently with death in the older compared with the younger patients. In OHCA-patients older than 70 years old exercise was no longer associated with prognosis (supplementary table). The longer time to ROSC the less was exercise found to relate to outcome.

The vast majority of the exercise-related OHCA occurred in men (96%), which has also been found in prior studies, but interestingly, recent studies have found that women are more active in sport and physical activity compared to men.^{18,19,30} The reason for the dramatic difference in incidence between the genders remains unknown.³² Nevertheless, the age and gender specific difference in prevalence of acute coronary syndrome, differences in the cardiac risk profile and a possible difference in the cause of acute coronary syndromes have been proposed as possible contributable causes.^{32,33}

Regular exercise has been proven effective for primary as well as secondary prevention of cardiovascular disease, but contradictory the risk of an acute cardiac event is temporarily increased during exercise – the so called paradox of exercise.^{1,34} In this study we found that the incidence of exercise-related OHCA was higher than the average proportion of daily time spend on exercising in the general population (7% vs. 2%/4%), which indicates a temporarily increased risk during exercise.^{30,31} Acute myocardial infarction is the most common cause of malignant arrhythmias and risk factors for exercise-related acute cardiac events such as prior cardiovascular events, smoking, and a recent episode of fatigue or flu-like symptoms have been associated with an increased risk.¹⁶ This study and in a few previous studies confirms acute coronary syndrome as a major contributor to OHCA occurring during exercise.^{2,16,19} Studies have discussed that the protective effect of exercise is more extensive than what can be attributed solely to changes in cardiovascular risk factors. It is evident that exercise improves vascular function and coronary collateral circulation. It has been suggested that the imbalance between oxygen supply and demand with the resulting ischemia during exercise combined with exposure to sympathetic activation could be protective in patients with regular exercise-habits in case of an acute cardiac event.^{18,35}

Most current studies in the field focus on sudden death in young professional athletes, but in our study only one out of 91 exercise-related OHCA-patients was a professional athlete.^{5–8} However, the rapidly growing segment of middle aged and senior athletes are important as the majority of events occur in this group with increased risk of both myocardial infarction and sudden cardiac arrest. Pre-participation screening of middle aged and senior athletes has been suggested as a strategy to improve survival in exercise-related cardiac arrest.^{7,8} In Denmark systematic pre-participation screening is not recommended, and the benefit of risk assessment in older citizens and patients with known cardiac disease are currently discussed.^{2,8,36,37}

Education of laypersons in CPR especially in fitness centers and sports arenas and use of AEDs together with a wider availability and strategic placement of AEDs in public areas have been suggested as a strategy to improve survival after sudden cardiac arrest in relation to exercise.^{11,38,39} According to the American Heart Association's guidelines exercise facilities have been identified as multiple cardiac arrest locations (likelihood of ≥ 1 cardiac arrest per 5 years), which corresponds well with our study where 43% of the exercise-related arrest occurred in gyms or sport stadiums.^{40–44} The use of

an AED by laypersons differs markedly in different countries with the Netherlands as an important role model with 36% of cardiac arrests treated with an AED, whereas the numbers are only 2% in Denmark and less than 1% in France.^{4,18,19}

There are some limitations to the present study. The study is retrospective and therefore only associations and not causalities can be derived. Patients with ACS might have a 'per se' better prognosis after OHCA compared to OHCA-patients with other etiologies, but such conclusions cannot be drawn from this study. Data are, however, consecutively collected based on the Utstein criteria. Unfortunately data were not available on exercise relation in patients not resuscitated from cardiac arrest or citizens found dead with obvious signs of death.⁸ Moreover symptoms prior to the cardiac arrest were not available. In estimating the risk of OHCA during sport compared to other activities we have compared the incidence of OHCA found in this study with the general population's hours spend on exercise. Some people, who stated that they perform strenuous exercise in the questionnaire, might suffer from comorbidities, which limit the strenuousness of the physical activity to a degree where cardiac arrest is unlikely. The comparison is an extrapolation and conclusions should carefully be drawn.

In conclusion, patients resuscitated from OHCA occurring during exercise have significantly lower mortality rates and acute coronary syndrome is significantly more common among exercise-related cardiac arrest patients. Even though cardiac arrest can occur during exercise our study shows that the prognosis and neurological outcome after OHCA is significantly better compared to patients suffering from arrest during other activities than exercise.

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Conflict of interest statement

None.

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